EXPERIMENT-06

SIMULATION-BASED INVESTIGATION OF MICROWAVE SEMICONDUCTOR DEVICES AND FREQUENCY-CONVERSION CHARACTERISTICS OF A MICROWAVE MIXER

PROGRAM

```
clc;
clear;
close all;
% Microwave Directional Coupler and Two-Way Power Divider
f = linspace(8,12,200);  % Frequency in GHz
f0 = 10;                 % Center frequency
% Generate S-parameters
S11 = -28 + 3*(1-((f-f0)/2).^2);
S21 = -0.8 - 0.2*((f-f0)/2).^2;
S31 = -10.5 - 1.5*((f-f0)/2).^2;
S41 = -30 + 3*((f-f0)/2).^2;
% Calculate performance parameters
Coupling = 10 + 2*((f-f0)/2).^2;
Isolation = 30 - 5*((f-f0)/2).^2;
InsertionLoss = 0.5 + 0.4*((f-f0)/2).^2;
Directivity = 20 - 6*((f-f0)/2).^2;
% Power divider parameters
DividerS11 = -26 + 1*((f-f0)/2).^2;
DividerS21 = -3.1 - 0.3*((f-f0)/2).^2;
DividerS31 = -3.1 - 0.3*((f-f0)/2).^2;
DividerIsolation = -30*ones(size(f));
PowerRatio = ones(size(f));
% Create figure
figure('Name','Microwave Directional Coupler and Two-Way Power Divider',...
    'NumberTitle','off',...
```

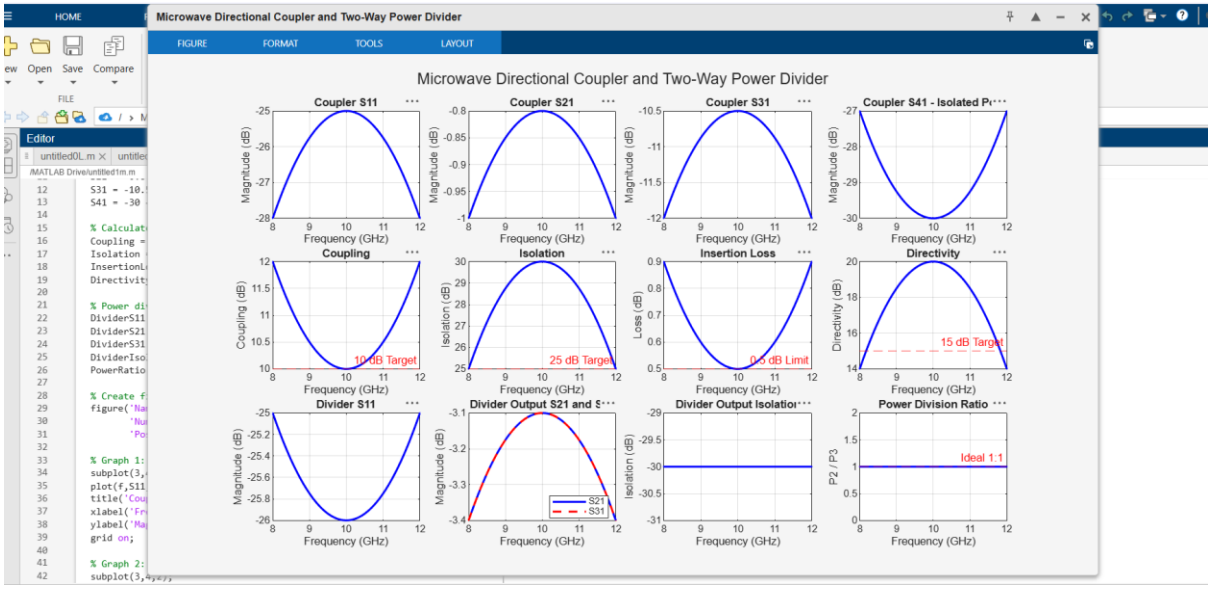
```
    'Position',[100 50 1200 750]);
% Graph 1: Coupler S11
subplot(3,4,1);
plot(f,S11,'b','LineWidth',1.8);
title('Coupler S11');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
grid on;
% Graph 2: Coupler S21
subplot(3,4,2);
plot(f,S21,'b','LineWidth',1.8);
title('Coupler S21');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
grid on;
% Graph 3: Coupler S31
subplot(3,4,3);
plot(f,S31,'b','LineWidth',1.8);
title('Coupler S31');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
grid on;
% Graph 4: Coupler S41
subplot(3,4,4);
plot(f,S41,'b','LineWidth',1.8);
title('Coupler S41 - Isolated Port');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
grid on;
```

```
% Graph 5: Coupling
subplot(3,4,5);
plot(f,Coupling,'b','LineWidth',1.8);
hold on;
yline(10,'r--','10 dB Target');
title('Coupling');
xlabel('Frequency (GHz)');
ylabel('Coupling (dB)');
grid on;
% Graph 6: Isolation
subplot(3,4,6);
plot(f,Isolation,'b','LineWidth',1.8);
hold on;
yline(25,'r--','25 dB Target');
title('Isolation');
xlabel('Frequency (GHz)');
ylabel('Isolation (dB)');
grid on;
% Graph 7: Insertion Loss
subplot(3,4,7);
plot(f,InsertionLoss,'b','LineWidth',1.8);
hold on;
yline(0.5,'r--','0.5 dB Limit');
title('Insertion Loss');
xlabel('Frequency (GHz)');
ylabel('Loss (dB)');
grid on;
% Graph 8: Directivity
subplot(3,4,8);
```

```
plot(f,Directivity,'b','LineWidth',1.8);
hold on;
yline(15,'r--','15 dB Target');
title('Directivity');
xlabel('Frequency (GHz)');
ylabel('Directivity (dB)');
grid on;
% Graph 9: Divider S11
subplot(3,4,9);
plot(f,DividerS11,'b','LineWidth',1.8);
title('Divider S11');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
grid on;
% Graph 10: Divider output S21 and S31
subplot(3,4,10);
plot(f,DividerS21,'b','LineWidth',1.8);
hold on;
plot(f,DividerS31,'r--','LineWidth',1.8);
title('Divider Output S21 and S31');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
legend('S21','S31','Location','best');
grid on;
% Graph 11: Divider Output Isolation
subplot(3,4,11);
plot(f,DividerIsolation,'b','LineWidth',1.8);
title('Divider Output Isolation');
xlabel('Frequency (GHz)');
```

```
ylabel('Isolation (dB)');
grid on;
% Graph 12: Power Division Ratio
subplot(3,4,12);
plot(f,PowerRatio,'b','LineWidth',1.8);
hold on;
yline(1,'r--','Ideal 1:1');
title('Power Division Ratio');
xlabel('Frequency (GHz)');
ylabel('P2 / P3');
grid on;
sgtitle('Microwave Directional Coupler and Two-Way Power Divider');
```

OUTPUT



RESULT

The simulation demonstrates diode behavior and microwave frequency conversion. For the input frequencies of 2.4 GHz and 2.3 GHz, the difference frequency is 100 MHz, and the sum frequency is 4.7 GHz.